

Did the Source Say That? A Source-Conditioned Per-Triple Faithfulness Gate for LLM Knowledge-Graph Construction

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Abstract—Large language models (LLMs) are now widely used to construct knowledge graphs by extracting (head, relation, tail) triples from text, but they hallucinate triples that are not supported by the source. We frame the problem as *per-triple faithfulness* (is the triple entailed by its source text), which is distinct from world truth, and introduce a learned, model-agnostic, source-conditioned faithfulness gate that scores each extracted triple against its evidence and prunes the unsupported ones. We build, to our knowledge, the first per-triple source-faithfulness benchmark, with a six-class taxonomy (a faithful class and five error classes, C1 to C5), derived from WebNLG, DocRED, REBEL, and SciERC. Our central finding is that the benefit of reading the source only appears when the negative triples are *plausible but absent*: with a naive (implausible) negative a strong entailment model is already near-ceiling without the source. On a hardened benchmark the source-conditioned gate reaches area under the ROC curve (AUROC) 0.996 [0.993, 0.999] versus 0.883 [0.858, 0.906] for an otherwise identical source-blind judge (paired bootstrap of the difference: 0.113 [0.091, 0.138]), detects 0.987 of plausible fabrications versus 0.295, and beats off-the-shelf NLI and LLM-as-judge baselines and a fine-tuned source-blind 7B judge (GraphJudge, 0.862); conditioning that same 7B (the 2x2’s missing cell) lifts it to 0.990, so conditioning, not scale, is the decisive factor. On a 475-triple human-labeled pilot (30 real unfaithful) of *real* LLM hallucinations the gate attains AUROC 0.899 (above the zero-shot baseline), although absolute recall on the rare real hallucinations is modest: we quantify this synthetic-to-real gap rather than hide it.

Index Terms—knowledge graph construction, faithfulness, hallucination detection, triple verification, natural language inference

I. INTRODUCTION

Knowledge graphs (KGs) store facts as triples and underpin search, question answering, and retrieval-augmented generation. Building them by hand is costly, so LLM-based triple extraction has become common. LLMs, however, hallucinate: they emit triples that read as plausible but are not stated in the source document, which corrupts the KG and propagates downstream.

We treat the filtering of extracted triples as a *faithfulness* problem. A triple is faithful if its source text supports it. Crucially, faithfulness is not the same as truth: a triple can be true in the world yet unfaithful to a source that does not state it, and for text-to-KG construction the right target is faithfulness. The closest prior systems each leave a gap. Judges that score a triple source-blind (for example GraphJudge [1]) accept a plausible but unstated triple; span-matching verifiers

(for example AEVS [2]) accept a triple whose entities are present but whose relation is wrong; holistic claim checkers do not operate per triple.

We contribute: (i) to our knowledge the first per-triple source-faithfulness benchmark with an error taxonomy, built from WebNLG, DocRED, REBEL, and SciERC, which isolates the plausible-but-absent failure mode; (ii) a learned, model-agnostic, source-conditioned gate, an entailment cross-encoder that is cheaper than LLM judges and beats off-the-shelf baselines; and (iii) an analysis showing that source-conditioning is the decisive factor, demonstrable only against negatives that are plausible-but-absent rather than merely implausible. The benchmark and the conditioning analysis are the contribution; the gate is deliberately the simplest sufficient instrument (a base-size, off-the-shelf NLI cross-encoder), so “it is just NLI fine-tuning” is a stated design choice, not a limitation.

II. RELATED WORK

LLM-based KG construction. LLMs increasingly extract entities and relations to assemble KGs from text, lowering cost but introducing hallucinated triples [11].

Per-triple verification. GraphJudge [1] fine-tunes an LLM (LLaMA2-7B with LoRA, low-rank adaptation) to judge triples, but judges each triple *source-blind*, so a plausible but unstated triple passes. AEVS [2] constrains extraction to source spans and verifies by matching, missing a wrong relation between present entities. PiVe [5] iteratively verifies a generated graph. Our gate is instead a learned, source-conditioned, per-triple scorer applied post-hoc to any extractor.

Holistic fact checking and NLI. GraphCheck [3] checks a whole claim against a source with a graph and a frozen LLM. Natural language inference (NLI)-based fact checking casts verification as entailment; we build on this with an NLI cross-encoder but target per-triple source-grounding. Fact-SelfCheck [6] and FActScore-style decomposition [7] detect hallucination at the fact level for free-text generation.

Contemporaneous KG benchmarks. FactCheck [12] validates world-truth of KG facts; KGHaluBench [13] probes LLM knowledge breadth; HalluGraph [14] and GraphEval [15] detect response-level hallucination by aligning source and response KGs. None provide a learned per-triple source-

TABLE I
FAITHFULNESS TAXONOMY WITH WORKED EXAMPLES, AGAINST THE SOURCE “AARHUS AIRPORT SERVES THE CITY OF AARHUS, DENMARK.”

Class	Type	Example triple
C0	faithful	(Aarhus Airport, city served, Aarhus)
C1	plausible, absent	(Aarhus Airport, operated by, US Air Force)
C2	wrong relation	(Aarhus Airport, leader party, Aarhus)
C3	entity error	(Billund Airport, city served, Aarhus)
C4	direction	(Aarhus, city served, Aarhus Airport)
C5	inference gap	needs joining two sentences

faithfulness score for construction or isolate the plausible-but-absent failure mode.

Filtering and process models. GraphRAG-FI [4] filters retrieved graph knowledge at QA time. Our gate is conceptually a step-level support scorer, related to process reward models [8], [9], [10], re-pointed from passage relevance to triple faithfulness.

III. PROBLEM FORMULATION AND TAXONOMY

Given source text D and a candidate triple $t = (h, r, o)$ from any extractor, the gate learns $\phi(t \mid D) \in [0, 1]$, approximately the probability that D entails t , and keeps t iff $\phi(t \mid D) \geq \tau$. The target is grounding in D , not world truth. We define a taxonomy of unfaithfulness (Table I): the faithful class C0 and five error classes, from plausible-but-absent fabrications (C1) to multi-sentence inference gaps (C5).

IV. BENCHMARK

We repurpose text-to-triple corpora loaded from Hugging Face parquet exports: WebNLG [16] and REBEL [17] (sentence-level), DocRED [18] (document-level, the source of C5), and the scientific corpus SciERC [19] (a distant domain for generalization). Positives are gold triples aligned to their source; negatives are controlled perturbations (C1 to C5).

Hardened C1 (the key design decision). A random fabrication is world-implausible, so a strong NLI model rejects it without reading the source, which hides the value of conditioning. We instead keep the gold relation and swap one argument for a type-consistent entity from that relation’s range, yielding a plausible-but-source-absent triple. The resampler rejects o' by surface form, so an alias of a present entity (for example “USAF” for “United States Air Force”) can occasionally slip through as a mislabeled negative, the main source of the residual C1/C2 noise bounded below. Splits are document-disjoint. An NLI-based label audit bounds accidental-faithful noise at about 9% for C1 and 17% for C2 (upper bounds); the high entailment the auditor reports for C4 is the NLI model’s direction-blindness, not label noise, which independently confirms the C4 finding below. The source corpora are WebNLG (CC BY-NC-SA 4.0), DocRED (MIT), and REBEL (CC BY-NC-SA 4.0); we release the derived

Algorithm 1 Hardened C1 (plausible-but-absent) negative

Require: gold triple (h, r, o) ; per-relation object pool O_r from the corpus

- 1: sample $o' \in O_r$ with $o' \neq o$ $\triangleright$ a type-consistent object for relation r
- 2: **while** the surface form of o' occurs in source D **do**
- 3: resample o' $\triangleright$ ensure it is genuinely absent
- 4: **end while**
- 5: **return** unfaithful triple (h, r, o')

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V. METHOD

The gate is a sequence-classification cross-encoder over (evidence, verbalized triple). The base is DeBERTa-v3-base [20] fine-tuned on NLI [21], since entailment pretraining captures argument order and semantic support. Evidence is selected by an entity-anchored multi-sentence selector (the sentences mentioning the head or the tail), which reduces to a single sentence for sentence-level corpora and supplies both entities’ context for document-level relations. We train with a class-weighted loss to counter the natural 1:4 faithful-to-unfaithful imbalance. The source-blind variant blanks the evidence and reproduces a GraphJudge-style judge, isolating the value of conditioning. The gate is robust to triple serialization: verbalized and linearized forms give AUROC 0.996 and 0.997; a DeBERTa-v3-large NLI base is no better (0.997), so we keep the cheaper base-size model. The score $\phi(t \mid D)$ is a probability, and the admission threshold τ is selected on development data for the desired precision-recall trade-off. Figure 1 shows the full pipeline.

VI. EXPERIMENTS

Synthetic test split (test_synth, 745 examples, 149 per class, classes C0 to C4; WebNLG is single-sentence and carries no C5, which is evaluated separately on DocRED in Section VI-E); DeBERTa-v3-base NLI base; 95% bootstrap confidence intervals over 1000 resamples, seed 42.

A. Source-conditioning is decisive

With the easy (implausible) C1, a source-blind model is already near-ceiling. With the hardened C1 the gap opens (Fig. 2): conditioned AUROC 0.996 [0.993, 0.999] versus blind 0.883 [0.858, 0.906], and C1 detection 0.987 versus 0.295. A paired bootstrap of the AUROC *difference* on aligned per-example scores gives 0.113 [0.091, 0.138] with $P(\Delta > 0) = 1.0$, so the advantage is significant.

B. Baselines and floors

The gate beats every baseline (Table II, Fig. 3). GraphJudge, a fine-tuned LLaMA2-7B judge, reaches only AUROC 0.862 source-blind, below our base-size conditioned gate (0.996) and even our own source-blind gate (0.883); a 7B cannot verify grounding it never sees, so it over-rejects faithful triples (C0

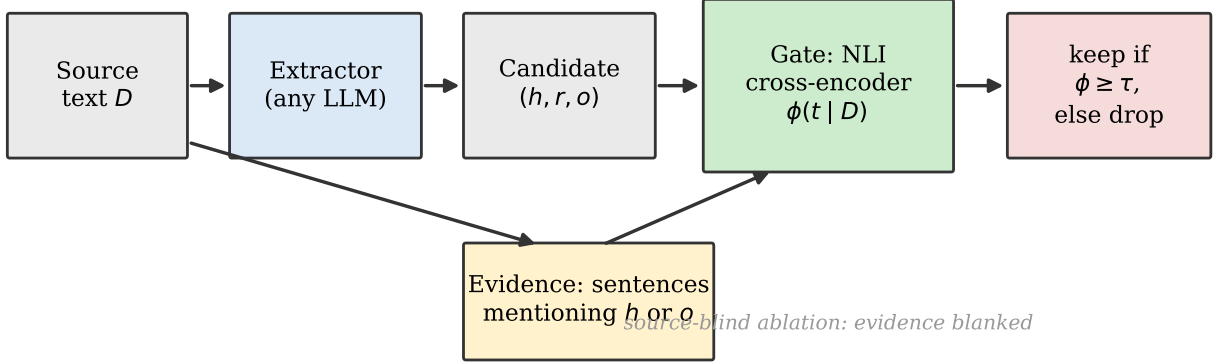


Fig. 1. The faithfulness gate. Any extractor proposes a triple (h, r, o) ; an entity-anchored selector retrieves the evidence sentences (those mentioning h or o); an NLI cross-encoder scores $\phi(t | D)$; triples scoring below τ are dropped. The source-blind ablation blanks the evidence, reducing the gate to a GraphJudge-style judge.

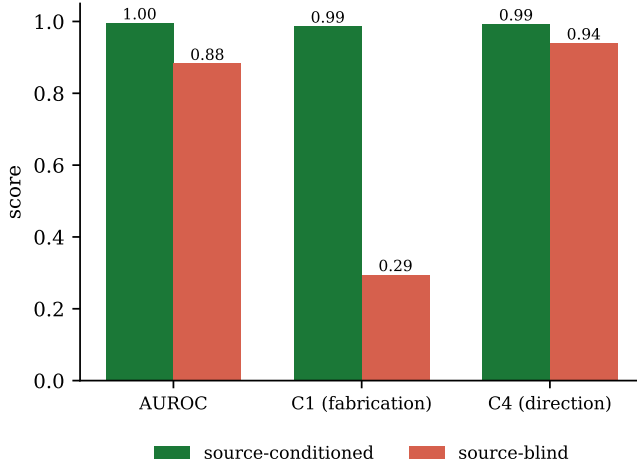


Fig. 2. Source-conditioned versus source-blind gate on the hardened benchmark: overall AUROC and per-class accuracy on the C1 (fabrication) and C4 (direction) classes. Conditioning is decisive on C1.

0.597). We reproduce GraphJudge’s defining property (a fine-tuned source-blind per-triple judge) on our benchmark, not its full pipeline (entity-centric denoising and instruction tuning); a full re-run is future work. Conditioning the same 7B judge (evidence in the prompt, all else fixed) raises it from 0.862 to 0.990, a large paired-bootstrap gain (+0.128 [0.098, 0.157], $P(\Delta > 0) = 1.0$). At fixed conditioning, the effect of scale is negligible in magnitude: the 184M base-size gate even edges the conditioned 7B by +0.007 [0.001, 0.013] ($P = 0.99$). This completes the 2×2 : conditioning, not parameter count, is the decisive factor, and the gate’s small residual edge is attributable to NLI-entailment pretraining (it is concentrated in the C4 direction class, 0.993 vs 0.852), not to size. The practical consequence is that a 184M cross-encoder matches a conditioned 7B at about 1/40 the parameters and, like the LLM

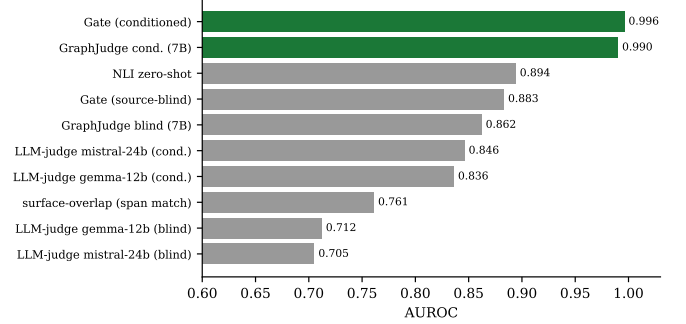


Fig. 3. AUROC on test_synth (higher bars at top). The two source-conditioned, fine-tuned methods (the gate, 0.996, and the conditioned 7B judge, 0.990) lead and are statistically tied (+0.007 [0.001, 0.013]); the source-blind methods trail.

judge, a small fraction of the inference cost (Section VI-C). Off-the-shelf source-conditioned methods (NLI zero-shot, the conditioned LLM-judge) catch C1 but fail C4 direction; only the gate, being both source-conditioned and fine-tuned on the taxonomy, handles both. The LLM-judge produces binary yes/no scores, so its AUROC is a single operating point and is not directly comparable to the continuous-score methods. A second, larger LLM judge (mistral-small:24b) confirms the pattern: conditioned AUROC 0.846 (C1 1.000 but C4 0.081) and blind 0.705, so off-the-shelf judges catch fabrications but not direction regardless of model size. As floors, a surface-overlap (span-matching) heuristic reaches only AUROC 0.761 and collapses on C2 (0.42) and C4 (0.18); always-keep and random sit at the 0.20 base rate and AUROC 0.53. This makes the advantage over span-matching verifiers empirical rather than definitional. The gate’s margin over NLI zero-shot (0.996 vs 0.894) largely reflects taxonomy fine-tuning; the claim-bearing comparison is the equally-trained source-blind gate (0.883), which isolates conditioning from training.

TABLE II
FAITHFULNESS DETECTION ON TEST_SYNTH (FULL SPLIT). BEST IN BOLD. THE LLM-JUDGE AUROC IS A SINGLE OPERATING POINT (BINARY SCORES), NOT DIRECTLY COMPARABLE.

Method	Source	Trained	AUROC	C1	C4
Gate (ours)	conditioned	yes	0.996	0.987	0.993
Gate (ours)	blind	yes	0.883	0.295	0.940
GraphJudge (LLaMA2-7B LoRA)	blind	yes	0.862	0.799	0.886
GraphJudge (LLaMA2-7B LoRA)	conditioned	yes	0.990	1.000	0.852
NLI zero-shot (DeBERTa-MNLI)	conditioned	no	0.894	0.933	0.107
LLM-judge (gemma3:12b)	conditioned	no	0.836*	0.980	0.074
LLM-judge (gemma3:12b)	blind	no	0.712*	0.537	0.262
LLM-judge (mistral-small:24b)	conditioned	no	0.846*	1.000	0.081
LLM-judge (mistral-small:24b)	blind	no	0.705*	0.785	0.550
Surface-overlap (span match)	conditioned	no	0.761	0.70	0.18

C. Cost

The gate scores 890 triples per second (1.1 ms per triple) on a single GPU, about 2300 times faster than the local LLM judge (2.6 s per triple); the autoregressive 7B judges (GraphJudge and the LLM-judge) share that per-triple generation cost, so the source-conditioning gain comes at negligible inference cost relative to any LLM-based judge.

D. Downstream gating

On the synthetic slice, which is constructed 1:4 so the keep-everything purity is only 0.20, gating raises the share of faithful kept triples to 0.973 at $\tau = 0.57$ while retaining 0.966 of faithful triples. That 0.20 base rate is an artifact of the construction, so we also report the realistic case: real extractor output is already about 94% faithful (Section VI-F), and there the same gate gives a smaller but real gain, lifting purity from 0.937 to about 0.96 at a few-percent cost in faithful retention. The dramatic synthetic figure is an upper bound; the realistic gain is modest.

E. Generalization to three unseen corpora

A gate trained only on WebNLG transfers to three unseen corpora (Fig. 4). On REBEL, a near-domain sentence-level corpus, it reaches AUROC 0.953 (C1 0.977). On the distant scientific domain of SciERC it degrades to AUROC 0.791: C1 (0.973) and C3 (0.984) still transfer, but the abstract relation types hurt C2, C4, and faithful recall, so generalization is strong near-domain and partial under heavy domain shift. On DocRED, unfaithfulness detection transfers, but faithful recall first collapses with single-sentence evidence because DocRED relations span sentences; entity-anchored multi-sentence evidence recovers it (faithful-recall progression 0.34 single-sentence to 0.59 cross-corpus to 0.93 in-corpus), and training in-corpus on DocRED reaches AUROC 0.992 with the C5 inference-gap class at 0.727 (on 11 test examples). The residual limitation is evidence selection, not the gate. On the sentence-level corpora (WebNLG, REBEL) the anchored selector returns the single source sentence, so it coincides with naive selection there; its benefit is specific to document-level relations, which justifies it as a default.

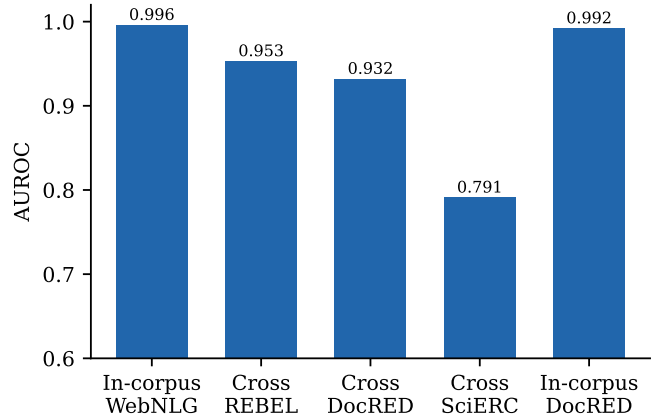


Fig. 4. Generalization AUROC: in-corpus WebNLG, cross-corpus REBEL, cross-corpus DocRED (multi-sentence evidence), cross-corpus SciERC (distant scientific domain), and in-corpus DocRED.

F. The natural slice

We evaluate on 475 human-labeled real triples (494 harvested from free local extractors on WebNLG; 19 uncertain dropped; 445 faithful, 30 unfaithful). The gate reaches AUROC 0.899 [0.853, 0.937], above NLI zero-shot (0.871), and at a fixed threshold ($\tau = 0.5$, recall 0.33) flags an order of magnitude more real hallucinations than that threshold on the blind judge (10 vs 1 of 30); mean gate scores separate the classes (0.965 faithful vs 0.688 unfaithful). The synthetic AUROC (0.996) overstates the real catch rate: with only 30 real unfaithful examples, absolute recall is modest and its interval wide, from 0.33 [0.17, 0.53] at $\tau = 0.5$ to 0.47 [0.28, 0.66] at $\tau = 0.95$ (exact binomial CIs). This is a small single-annotator pilot, not full validation; we report it to quantify the synthetic-to-real gap that synthetic-only evaluation hides.

Calibration, AUPRC, and threshold transfer. The gate is well calibrated: expected calibration error is 0.011 on test_synth and 0.067 on the natural slice (Fig. 6), so ϕ is a usable probability. Under area under the precision-recall curve (AUPRC) on the rare unfaithful class (base rate 0.063) the picture is humbler: the gate scores 0.326 and NLI zero-shot 0.393, so on real minority detection neither is strong and

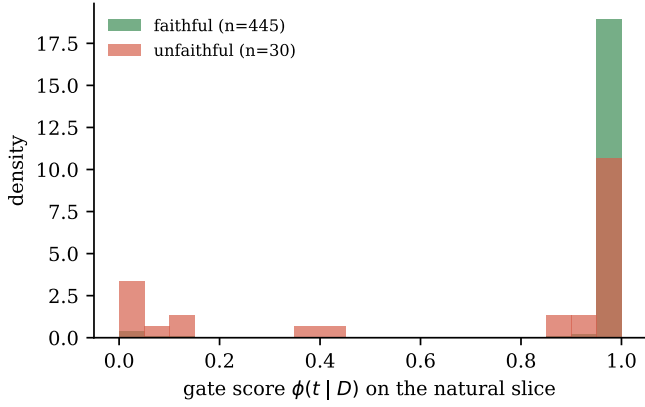


Fig. 5. Gate scores on the natural slice: faithful versus the 30 real unfaithful triples. The means separate (0.965 vs 0.688), but the unfaithful tail overlaps the faithful mass, which is why fixed-threshold recall is modest.

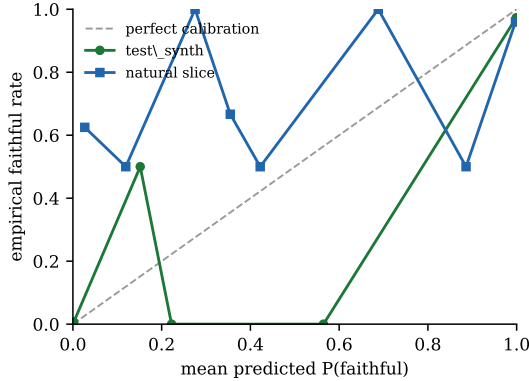


Fig. 6. Reliability diagram: predicted $P(\text{faithful})$ vs the empirical faithful rate. Well calibrated on test_synth (ECE 0.011), reasonably on the natural slice (ECE 0.067).

AUROC (0.899 vs 0.871) flatters both. The mechanism of the synthetic-to-real gap is *threshold transfer*: real unfaithful triples have mean score 0.688 and rank correctly (AUROC 0.899), but the synthetic-calibrated τ does not transfer (recall 0.33 at $\tau = 0.5$); in practice τ should be set on a small in-domain labeled sample.

G. Error and overlap analysis

Aligning the learned methods per example (Table III), of the 149 C1 examples the conditioned gate catches 103 that the source-blind gate accepts, and zero the other way, so source-conditioning strictly dominates on plausible fabrications. Only one unfaithful triple is missed by all five methods. Two cases make the taxonomy concrete. For C1, the triple (Aarhus Airport, operating organisation, United States Air Force) against a source naming a different operator is a plausible-but-absent fabrication: the conditioned gate scores it 0.00 (rejects) while the source-blind gate scores 0.71 (accepts). For C4, the inverted triple (Aarhus, Denmark; city served; Aarhus Airport), whose source states the reverse direction, has both entities present: the NLI-fine-tuned gate still rejects it (C4

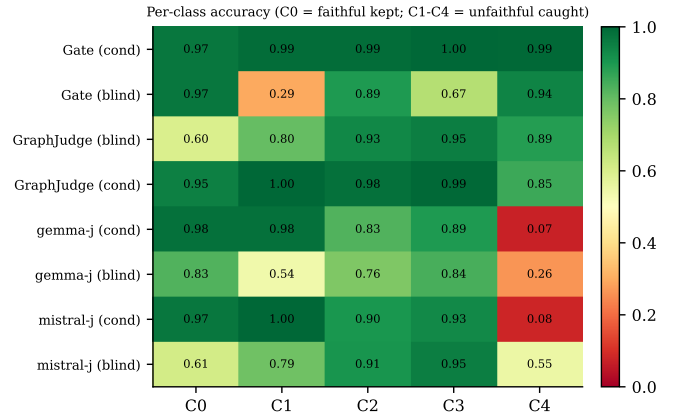


Fig. 7. Per-class accuracy by method (C0 = faithful kept, C1-C4 = unfaithful caught). The conditioned gate is the only method strong across the whole row.

TABLE III
PER-CLASS ACCURACY BY METHOD (TEST_SYNTH). G-C/G-B: GATE CONDITIONED/BLIND; GJ-B/GJ-C: GRAPHJUDGE BLIND/CONDITIONED; LJ-C/LJ-B: LLM-JUDGE CONDITIONED/BLIND.

Class	g-c	g-b	GJ-b	GJ-c	LJ-c	LJ-b
C0	0.966	0.966	0.597	0.946	0.980	0.826
C1	0.987	0.295	0.799	1.000	0.980	0.537
C2	0.987	0.893	0.933	0.980	0.826	0.758
C3	1.000	0.671	0.953	0.993	0.893	0.839
C4	0.993	0.940	0.886	0.852	0.074	0.262
all	0.987	0.753	0.834	0.954	0.750	0.644

accuracy 0.993) where a relevance model accepts it. The two classes need different capabilities (reading the source for C1, argument order for C4), and only the gate has both.

VII. LIMITATIONS

The synthetic negatives are constructed; the human-labeled natural slice (AUROC 0.899) is a small, single-annotator pilot (30 real unfaithful examples) and reveals a synthetic-to-real gap (0.996 to 0.899, with modest absolute recall). Inter-annotator agreement and a larger harvest are future work. The dramatic synthetic downstream purity gain (0.20 to 0.97) is an artifact of the 1:4 construction; the realistic gain is modest (0.937 to about 0.96). Document-level corpora need multi-sentence or provenance-aware evidence, and evidence selection is part of the deployed system, not a disclaimable preprocessing step. The LLM-judge baseline uses binary scores (single operating point). Results use a base-size model; larger bases may shift absolute numbers. GraphJUDGE is reproduced as a source-blind LLaMA2-7B LoRA judge, not its full pipeline; the C5 class is evaluated on few examples (11).

VIII. CONCLUSION

Filtering LLM-extracted triples by source faithfulness, with a learned source-conditioned gate, substantially improves knowledge-graph purity on constructed data and gives a smaller real-world gain. The benefit of reading the source is

real and significant, but only measurable against plausible-but-absent negatives, which is the central lesson for benchmarking faithfulness; and the synthetic-to-real gap we measure is a caution for the many synthetic-only benchmarks in this space.

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